AI360

TEAM

14

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PAPER

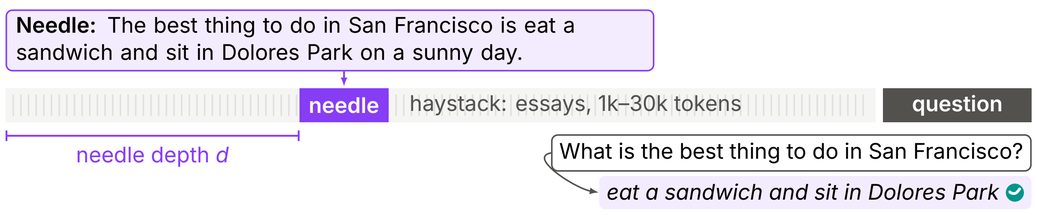
Retrieval Head Mechanistically Explains Long-Context Factuality

Wu et al., 2024 · arXiv:2404.15574

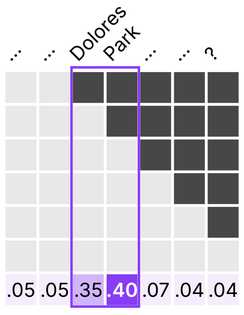
01 THE PAPER

Which attention heads pull a fact out of a long context?

Wu et al. [1]: a few **retrieval heads** copy facts from context; masking them breaks Needle-in-a-Haystack [2]. We reproduce (03) and stress-test this (04–07).



02 HOW WE SCORE A HEAD



One head, one decoding step. Generated token: **Park**

retrieval score (paper)
Top-attended token is in the needle N and equals the generated token.

$$\mathrm{RS}_h = \frac{\#\,\text{copied}}{|N|}$$

attention mass (ours)
Attention on N at answer steps D that emit a needle token.

$$\mathrm{AM}_h = \frac{1}{|D|}\sum_{t\in D}\sum_{i\in N} A_{t,i}$$

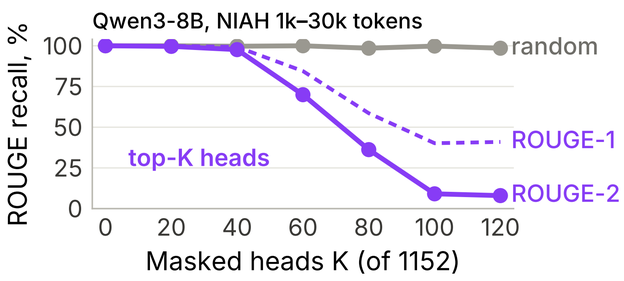
QRScore [3]
Attention from question tokens Q to N, before answering.

$$\mathrm{QR}_h = \frac{1}{|Q|}\sum_{q\in Q}\sum_{i\in N} A_{q,i}$$

Masking test. A masked head attends uniformly while the answer is decoded. Top-K heads of a score vs random or bottom-K heads.

03 REPRODUCTION

Exact check. The authors' code on Qwen1.5-14B-Chat finds the paper's top-20 heads (20/20); our code matches it ($r = 0.994$, same top-50).



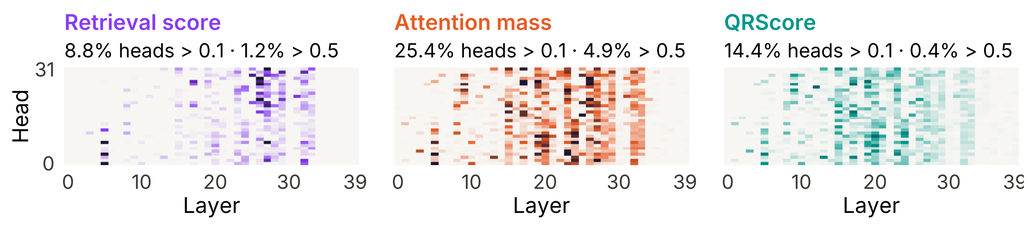
5.4%
of 1152 heads score > 0.1

100 → 8
ROUGE-2, top 120 heads masked (120 random: 98.5)

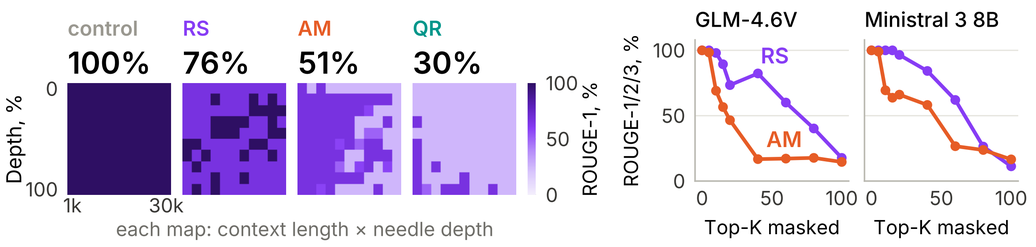
Result. Reproduced with the paper's **retrieval score**: a sparse set of heads is necessary; the answer is lost, not paraphrased (6 of 3800).

04 WHICH SCORE WORKS BEST?

GLM-4.6V-Flash: each score ranks all 1280 heads on 27 NIAH prompts.



Top-20 heads of each score masked (control: 20 lowest); right: more heads.

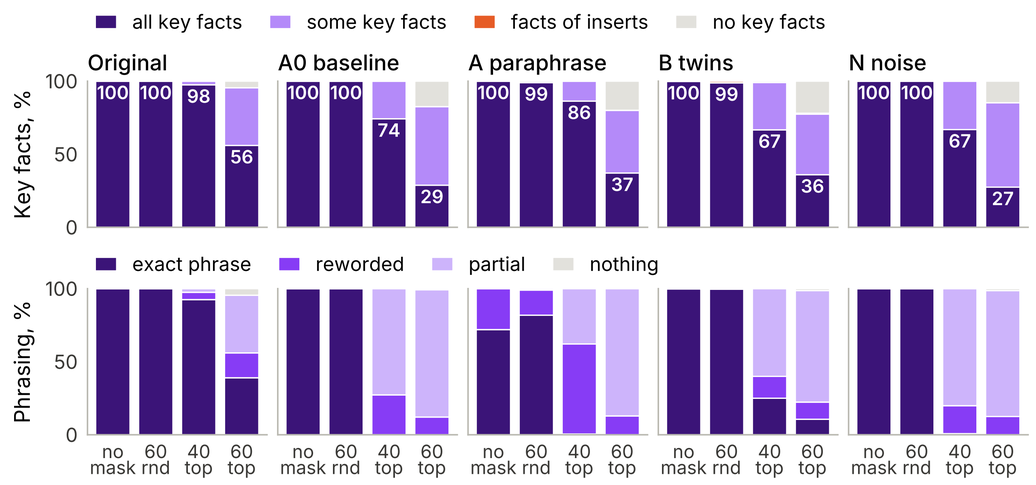


Result. Attention-based scores find the critical heads: top-20 by AM or QR cut recall to 51% and 30%, by RS only to 76%; on both models 10 AM heads already drop it to 69%. AM and QR correlate ($\rho = 0.96$) yet share only 7 of their top-20.

05 HARDER RETRIEVAL

Needle	Every visitor to the **Lisbon** Tram Museum is asked to ring the brass bell at the entrance and sign the guest book with a green pen.
Twin (B)	Every visitor to the **Porto** Tram Museum is asked to pull the iron lever at the gate and stamp the entry card with a red seal.
Question	What is every visitor to the Lisbon Tram Museum asked to do?

- No mask — "…ring the brass bell at the entrance and sign the guest book with a green pen."
- 60 random heads — same facts, after an introductory phrase
- 60 top heads — "…asked to **pull the lever and stamp the entry card.**" Porto's facts, told about Lisbon.

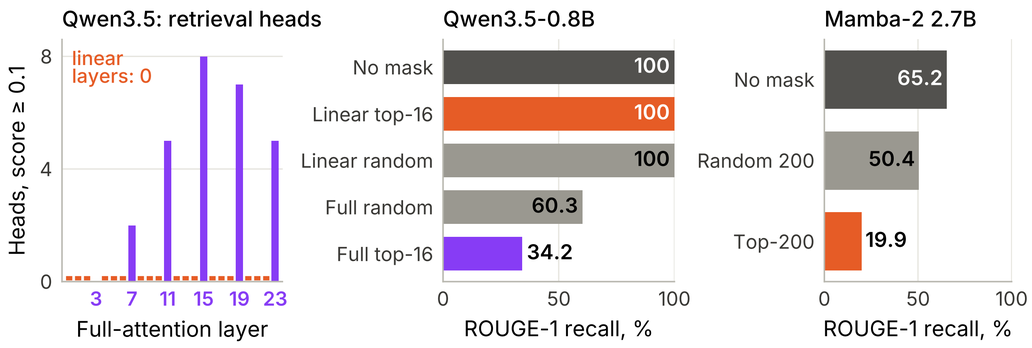


A0: new needles, plain question. A, B, N: levels A1–A3, B1–B4, N1–N4 pooled.

Result. The simple-task heads matter on every needle: with 60 masked, 27–56% of answers keep all key facts (99–100% for 60 random). The drop comes mostly from the new needles (A0: 29%); twists A, B, N shift it by at most 12 points. Wording breaks first: at 40 heads exact phrases nearly vanish, while 67–86% of answers keep all facts.

06 LINEAR AND HYBRID ATTENTION

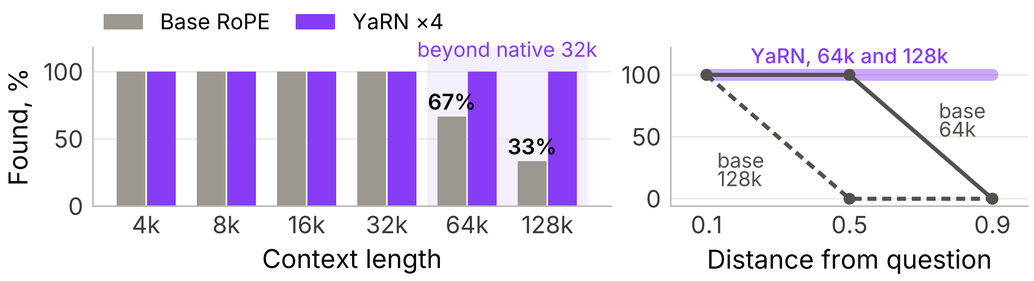
Linear attention has no softmax map, so a head is scored by token contributions $|\alpha_{t,s}|\,\|v_s\|$ [4]. Qwen3.5 mixes 18 linear and 6 full-attention layers. Mamba-2 [5] is purely linear; its heads are masked by freezing the state update ($\Delta_h = 0$).



Result. The hybrid model retrieves only through full attention; its linear heads do not matter. Pure-linear Mamba-2 still grows retrieval heads (2.2%), 71% of them in the first 12 of 64 layers.

07 BEYOND THE CONTEXT WINDOW

Qwen2.5-7B-Instruct (4-bit, native window 32k). Only the RoPE scaling changes: base vs YaRN [6], factor 4.



Result. YaRN removes the failure past 32k (128k: 33% → 100%); base fails first on far needles. 3 prompts per length past 32k.

References [1] W. Wu et al. "Retrieval Head Mechanistically Explains Long-Context Factuality". *arXiv:2404.15574* (2024). [2] G. Kamradt. *Needle In A Haystack: Pressure Testing LLMs*. GitHub. 2023. [3] W. Zhang et al. "Query-Focused Retrieval Heads Improve Long-Context Reasoning and Re-ranking". *EMNLP*. 2025. [4] G. Kobayashi et al. "Attention is Not Only a Weight: Analyzing Transformers with Vector Norms". *EMNLP*. 2020. [5] T. Dao et al. "Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality". *ICML*. 2024. [6] B. Peng et al. "YaRN: Efficient Context Window Extension of Large Language Models". *arXiv:2309.00071* (2023). [7] C.-P. Hsieh et al. "RULER: What's the Real Context Size of Your Long-Context Language Models?" *arXiv:2404.06654* (2024). [8] Y. Lin et al. "Retrieval Heads are Dynamic". *ACL*. 2026. [9] X. Zhao et al. "Understanding Synthetic Context Extension via Retrieval Heads". *ICML*. 2025. [10] Y. Gao et al. "U-NIAH: Unified RAG and LLM Evaluation for Long Context Needle-In-A-Haystack". *arXiv:2503.00353* (2025).